AARADHYA BHATIYA

Game Developer · Technical Artist · Game Designer

Unity · Roblox (Luau) · HLSL shaders · Gameplay systems · Systems & balance design

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Portfolio | LinkedIn | GitHub | itch.io

Summary

Game developer, technical artist and game designer with 3+ years building shipped games and real-time applications across mobile, WebGL, AR and Roblox. I take games from GDD to release: gameplay architecture and data-driven systems, custom HLSL shaders and visual development, and systems design backed by numerical balancing. Recently I owned the full Unity client of a 3D companion app, lead a five-person team on an original horror game, and designed and built a large co-op Roblox game end-to-end with an AI-assisted workflow.

Skills

Languages	C#, Luau/Lua, C/C++, HLSL, GML, Python, JavaScript, SQL
Engines	Unity 3D/2D (URP), Roblox Studio, GameMaker Studio
Gameplay & Systems	Behaviour and state architecture, animation systems and state machines, character controllers, physics, AI pathfinding, server-authoritative multiplayer (Netcode for GameObjects, Roblox), data-driven content pipelines, custom editor tooling
Technical Art	Custom HLSL shaders, Shader Graph, real-time VFX, volumetric fog and lighting, planar reflections, procedural generation, OpenGL 3.3, WebGL, profiling and optimisation, Blender, Aseprite, pixel art
Game Design	Systems and progression design, economy and numerical balancing (RTP and probability models, Monte Carlo simulation), difficulty tuning, encounter and pacing design, GDDs, playtest-driven iteration
Roblox Platform	DataStore (session-locked saves), MemoryStore, TeleportService and reserved servers, MarketplaceService, voice and audio API
Pipeline & Tools	Android/iOS builds, Play Store submission, IronSource, analytics, native plugins (Objective-C, Java/JNI), React Native interop, cross-device QA, AI-assisted development (Claude Code, Roblox Studio MCP), Git/GitHub, JIRA, Asana, Figma

Experience

Unity Developer Boop | Remote, contract — Jun 2026 – Sep 2026

Unity (URP), C#, iOS and Android Showcase video · heyboop.ai

- **Sole owner of the Unity client** for a 3D character-driven companion app after the other Unity developers departed, delivering all gameplay, animation, rendering and tooling work through to a near-release build.
- **Data-driven behaviour system:** 27 ScriptableObject behaviours selected from live context by requirement matching and a priority ladder, with recency-weighted random selection so ambient behaviour reads as spontaneous, not looping.
- **20-step execution interpreter** composing performances from reusable steps (anchor navigation, animation playback, gesture-scrubbed clips, rig displacement, camera control), so behaviours are authored in the Inspector without code; plus 11 custom editor tools.
- **Animation and movement:** displaced the collider-free visual rig to interaction anchors so furniture animations play correctly while physics stays legal; moved movement onto a single-writer motor, fixing a Rigidbody/transform desync that caused drift; built animation variant swapping, spring-bone motion and frame-accurate audio cues locked to the animator playhead.
- **Performance:** three URP quality tiers with automatic device-tier detection and a zero-allocation CPU/GPU profiling overlay; on-device profiling showed the engine used only 6–11 ms of a 33 ms frame budget, redirecting work to main-thread contention.
- **Platform integration:** designed and documented the Unity ↔ React Native message contract (acknowledgement protocol, two-phase handshake that removed a startup deadlock); native plugins in Objective-C and Java/JNI with tiered degradation across device capability levels.

Lead Game Developer TONIZTOZ | Remote — Mar 2025 – Present

The God They Buried · Unity 2D, C#, HLSL, Aseprite Demo video · toniztoz.com

- **Sole senior developer leading a five-person cross-disciplinary team** on an original 2D pixel-art horror game rooted in Asian folklore; own it from storyline, storyboard and GDD through active development, and mentor juniors on Unity, Aseprite and asset workflows.
- **Technical direction and architecture:** full project groundwork and core systems end-to-end, including the player animation pipeline and player class architecture.
- **All visual development in custom HLSL:** 2D volumetric fog, a pixel-art horror lighting pipeline, and bespoke ghost and demon shaders with distortion and translucency.

Unity Developer Intern Pythrust Technologies | Gurugram, India Oct 2025 – Jan 2026

2MinWin by MINWIN Digital Studios · Unity, C# 2min.win

- **Built 17+ complete mini-games from scratch** for a live production title in a five-person team, implementing multiplayer, VFX, UI and SFX end-to-end for each, and working with UI and sound design to keep proposed designs replicable in engine.
- **Integrated the IronSource ad mediation SDK** alongside analytics and backend services; ran cross-device regression testing and tracked delivery in Asana.

Independent Contractor Freelance | Remote 2024 – Present

Unity, C#, WebGL, AR Foundation Casino showcase · AR / VR showreel

- **Four real-time casino games** (Mines, Roulette, Crash, Teen Patti) shipped as Unity HTML5 builds inside a client's web platform: designed the RTP mathematical models and probability systems, live wallet integration with balance sync and currency switching, and retention-focused gameplay loops.
- **15+ AR Unity projects** for clients across desktop and mobile: real-time tracking systems, indoor navigation built on 3D spatial scans, and interactive AR environments.

Projects

Paranormal Inc. Co-op horror investigation game 2026 – Present

Roblox, Luau · Designer and developer

- **Designed and built a 1–4 player first-person investigation game:** 4 maps (80 rooms), 5 entities with a 7-type evidence system, 10 upgradeable devices, 45 paranormal events and 12 location stories.
- **Systems and balance:** 5 difficulty tiers built from separate levers, an itemized economy with one-time rewards and penalties, and a 16-rank progression curve tuned with Monte Carlo simulation; an agitation-based pacing model drives the entity AI and event cadence.
- **Multiplayer and live-service:** lobby-to-crew-server teleports with host and voting flow, session-locked DataStore saves, MemoryStore private-server listings, voice-driven mechanics (mic loudness, speech-to-text), and no-pay-to-win developer products.
- **Architecture:** server-authoritative, ~43k lines of Luau (38 services, 40 client controllers, 15 data-definition modules), built with an AI-assisted workflow (Claude via Roblox Studio MCP) and 25 in-house editor tools; every system reviewed and playtested, including client-side snapshot interpolation for smooth AI movement.

Chalo Yaar! 3D multiplayer travel game Jan 2026 – May 2026

Unity (URP), C#, HLSL, Netcode for GameObjects itch.io · Trailer

- **Cozy first-person road-trip game:** drivable campervan hub, wildlife photography progression, inventory and survival stat systems on a modular multiplayer architecture.
- **Custom HLSL rendering:** procedural water with Worley-noise caustics, volumetric fog from 3D noise, and GPU foliage wind via vertex displacement.
- **Optimisation:** cut draw-call batches from 130,000+ to under 30,000 and raised sustained framerate from 15 to 50+ fps through shadow caster reduction, LOD tuning and fog-masked foliage culling.

SlenderAR AR horror survival 2025

Unity, AR Foundation, Android itch.io · Trailer

- **The Slenderman experience in AR:** portal-based entry from the real world into a haunted forest, real-world walking locomotion, player-tracking AI and dynamic audio. Free on itch.io.

Lucius: The Endless Maze Runner 3D horror maze runner 2024 – 2025

Gameplay video

- **Procedural levels** from a backtracking algorithm, with enemy pathfinding that adapts to the player's position through each generated maze.

Other work

- **Himeji Castle** (Blender): environment piece with particle-driven sakura petals, wind physics and bokeh camera work.
- **AssistSense** (Python, PyQt6, two-person team): macro engine with timed input recording, IF/ELIF/ELSE playback and a wake-word voice listener.
- **Questify** (C#): gamified task manager that verifies real productivity through a GraphQL API and GitHub activity tracking.

Education

B.Tech Computer Science and Engineering (spz. Gaming and Graphics) UPES, Dehradun 2022 – 2026

CGPA 7.78